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Modeling biopore effects on root growth and biomass production on soils with pronounced sub-soil clay accumulation

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ABSTRACT

Soils with subsoil clay accumulation account for more than 20% of the global land surface. These soils are characterized by vertical differences with respect to soil texture and increasing bulk density below the topsoil, which in turn affects root penetration into the subsoil. Biopores are preferential pathways for roots and assist in overcoming physical barriers like high density soil layers. An integration of these relationships into cropping systems models at the field scale is on-going. This paper presents a new approach to model the effect of biopores on root development in soils with clay accumulation at the plot scale. In this approach, the effect of biopores on root elongation rate depends on bulk density and on a biopore-root growth threshold (MPRT), which is the biopore volume at which the resistance of soil strength to root penetration is completely offset by the density of the biopores. The approach was integrated into a model solution of the model framework SIMPLACE (**S**cientific **I**mpact assessment and **M**odeling **PL**atform for **A**dvanced **C**rop and **E**cosystem management). MPRT was parameterized for spring wheat using the inverse modeling approach based on root observations from a multi-factorial field experiment on a Haplic Luvisol. The observed biopore densities (>2 mm diameter) were between 300 and 660 pores m^{-2} (equivalent to a volumetric proportion of 0.38–0.83%) depending on the preceding crop. Observed soil bulk densities ranged between 1.31 and 1.62 $g\,cm^{-3}$. For spring wheat, the best fit between simulated and observed root densities in different layers was obtained with a MPRT of 0.023 $m^3\,m^{-3}$ (equivalent to 2.3% of soil volume). The mean simulated total above ground biomass was sensitive to MPRT and had the best agreement with observed values when a MPRT between 0.023 and 0.032 $m^3\,m^{-3}$ was used in the simulations. Scenario simulations with the parameterized model at the same site demonstrate the importance of biopores for biomass production of short-cycle spring wheat when prolonged dry spells occur. The simulations allow a rough quantification of the biopore effects with respect to root elongation rate and biomass production at the plot scale with the potential to be extended to the field scale.

1. Introduction

In recent decades, significant progress has been made in the modeling of crops including root growth. The focus of many modeling efforts was on simulating the response of crops to diverse climate, soil and management conditions, e.g. CropSyst (Stockle et al., 2003), DSSAT (Jones et al., 2003), APSIM (Keating et al., 2003a), STICS (Brisson et al., 2003a) and the different models developed at Wageningen University (van Ittersum et al., 2003). However, most models consider effects of soil structural properties in a simplified way, using mainly soil bulk density as a proxy for penetrability of the soil layers (Keating et al., 2003b; Williams and Izaurralde, 2005). Soil bulk density affects simulated root penetration rates and in some cases extension of lateral roots (Brisson et al., 2003b). In soils with little vertical differentiation of bulk density and low macropore density, this simplification may yield satisfactory results when simulating root growth and development. However, in soils with pronounced vertical differences with respect to soil structure and macropore density, such simplified assumptions may lead to uncertain rates of root growth and of water and nutrient uptake. This is due to the fact that macropores generated by roots do not reduce soil bulk density, because the reduction of density in the voids created by the roots is largely compensated by the increased soil density of the root channel walls (Hirth et al., 2005; Young, 1998). Vertical differences with respect to soil texture and bulk density are characteristic features of soil types with clay accumulation

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like Luvisols, Alisols, Acrisols, Lixisols, Albeluvisols, and Nitisols. Besides, soils belonging to the soil group Planosols, are characterized by abrupt texture changes within the soil profile, mostly due to lithological heterogeneities. These seven soil groups together account for more than 20% of the global land surface (FAO, 2003). Luvisols, and in particular those derived from loess, are among the most intensively cropped soils in Central, Southern and Eastern Europe (FAO, 2003). Therefore, the influence of soil structure (porosity, pore size distribution) on root distribution patterns of food crops needs to be investigated and modeled (Wang and Smith, 2004).

Biopores are round-shaped biopores created by soil animals and decaying plant roots (Hirth et al., 2005). The importance of biopores as preferential pathways for roots to overcome physical barriers like high density soil layers has frequently been stressed (Stirzaker et al., 1996b; Volkmar, 1996). However, little work has been done to integrate such responses into plant and crop models. In some cases biopore effects on root growth and soil water dynamics have been investigated at the pot scale (Stirzaker et al., 1996a) or integrated into soil water balance models (Bronstert and Plate, 1997; Jakobsen and Dexter, 1987) but integration of these relationships into cropping systems models and model testing at the plot and field scale is on-going. Wu et al. (2007) developed a field scale model which combines model components for plant growth and development, nitrogen and carbon cycling as well as soil water dynamics with a three-dimensional root system sub-model as an interface. However, the interaction between soil structure and root development follows the simplified approach of other field scale crop models.

In this study, an attempt is made to model the effect of biopores on root development in a soil with subsoil clay accumulation. A new approach is proposed which is applicable at the plot scale with the potential to be extended to the field level. The approach has been integrated into a crop model solution of the model framework SIMPLACE (**S**cientific **I**mpact assessment and **M**odeling **PL**atform for **A**dvanced **C**rop and **E**cosystem management). We first describe the modeling approach and the implemented crop model solution in SIMPLACE as used in this study. Then, we present the parameter optimization procedure for spring wheat at the plot scale, using inverse modeling based on root observations from a multi-factorial field experiment in Germany. Finally, we apply the calibrated model, to demonstrate, via a sensitivity analysis, the importance of biopores for root growth, water and nutrient uptake as well as biomass production of drought stressed spring wheat.

2. Model development

2.1. The modeling framework SIMPLACE

SIMPLACE (**S**cientific **I**mpact assessment and **M**odeling **PL**atform for **A**dvanced **C**rop and **E**cosystem management) is a modeling framework which is based on the concept of encapsulating the solution of a modeling problem in discrete, replaceable, and interchangeable software units called SimComponents or sub-models (Enders et al., 2010). This structure of the model allows for an easy plug-in of new sub-models and the use of different approaches for the simulation of processes via alternate mathematical formulations. Presently, SIMPLACE contains sub-models for the major processes affecting crop growth (including root growth), i.e. (1) crop phenology and development, (2) shoot growth, (3) crop nitrogen demand, (4) crop water demand, (5) root growth, (6) soil temperature dynamics, (7) soil water dynamics and root water uptake, (8) soil carbon and nitrogen turnover and (9) soil mineral nitrogen dynamics (N uptake and N leaching) (Table 1). In addition, a flexible vertical soil discretization was implemented, allowing to accounting for different pore systems in the individual soil layers. For the current model application a layer thickness of 0.03 m was chosen as recommended by (Addiscott and Whitmore, 1991).

2.2. The model solution

A specific combination of sub-models within the framework is a model solution. For the purpose of this paper the solution SIMPLACE<Lintul2,STMPsim,SlimRoots,Hillflow,SoilCN> has been used, consisting of the combination of the sub-models Lintul2, SLIMRoots, Hillflow and SoilCN (Fig. 1 and Table 1). In this solution of SIMPLACE crop growth and development is solved by the LINTUL2 sub-model (van Oijen and Lefelaar, 2008), which has frequently been used and validated (Farre et al., 2000; Gimplinger and Kaul, 2009; Shibu et al., 2010; Wolf et al., 2002). In LINTUL2, crop development is driven by mean air temperature and crop specific base temperature. Carbon assimilation and crop growth rate depend on photosynthetic active radiation, leaf area index (LAI) and crop specific light use efficiency (LUE) according to the following equation:

$$\mathrm{GTOTAL} = \mathrm{LUE} \times \mathrm{PARINT}$$

where GTOTAL is the potential daily crop growth rate ($\mathrm{g\,m^{-2}\,d^{-1}}$) and PARINT is the intercepted photosynthetic active radiation ($\mathrm{MJ\,m^{-2}}$).

GTOTAL is reduced when root available soil water does not meet crop transpirational demand. In addition, nitrogen stress is considered as a growth constraint and affects LAI development, specific leaf area (and hence PARINT) as well as light use efficiency according to the approach used in LINTUL3 (Shibu et al., 2010). The assimilates are distributed to plant organs (leaves, stem, storage organs and roots) according to crop and development specific partitioning tables (Angulo et al., 2013; van Oijen and Lefelaar, 2008).

The soil in this SIMPLACE solution is subdivided into 50 layers with a thickness of 0.03 m each. In the SlimRoots sub-model, root growth is separated into the growth of seminal and lateral roots (Addiscott and Whitmore, 1991). In the original approach, maximum growth of seminals (ASROOTMAX in $\mathrm{g\,m^{-2}\,d^{-1}}$) is limited by the specific weight of seminals (WSROOT in $\mathrm{g\,m^{-1}}$) and a maximum penetration rate SROOTRATEMAX being set to 0.015 $\mathrm{g\,m^{-1}}$ and 0.033 $\mathrm{m\,d^{-1}}$ respectively for summer wheat (Bingham and Wu, 2011; Jamieson and Ewert, 1999; Pritchard et al., 1987):

$$\mathrm{ASROOTMAX} = \mathrm{SROOTRATEMAX} \times \mathrm{WSROOT} \times \mathrm{SROOTN}$$

with SROOTN being the number of seminals per squaremeter ($\mathrm{m^{-2}}$). The actual seminal growth (ASROOT) is limited by the supply of assimilates from the shoot (RWRT in $\mathrm{g\,m^{-2}}$) and the root penetration velocity related to the soil temperature in the deepest soil layer with roots (TCRATE in $\mathrm{m\,d^{-1}}$) according to (Jamieson and Ewert, 1999) with

$$\mathrm{ASROOT} = \mathrm{TCRATE} \times \mathrm{WSROOT} \times \mathrm{SROOTN}$$

The growth of lateral roots in the layers with seminal roots is determined by the soil water content in each layer and the assimilates (RWRT) provided by the shoot (minus assimilate consumption ASROOT by the seminals), establishing a feedback between soil available water and nitrogen on the one hand and shoot and root growth on the other hand. In order to take into account vertical differences in soil structure especially in soils with clay accumulation like Luvisols, the effect of soil strength on root growth has been incorporated into the SlimRoots sub-model (see next section).

Soil temperature fluctuation in each soil layer depends on the soil surface temperature (including the effect of snow cover and crop cover), the distance of the layer from the soil surface and the damping depth, where soil temperature is equal to the annual

Table 1
Sub-models selected for the development of the SIMPLACE model solution describing the most relevant processes for crop growth, water and nutrient uptake in the field trial.

Soil/plant	Process	Model/source
Soil	Water fluxes (physically based)	HILLFLOW 1D (Bronstert and Plate, 1997)
	Water fluxes (conceptual)	SlimWater (Addiscott et al., 1986; Addiscott and Whitmore, 1991)
	Soil temperature	STMPsim (Williams and Izaurralde, 2005)
	C/N turnover	SoilCN (Corbeels et al., 2005)
	Mineral N leaching and uptake	SlimNitrogen (Addiscott and Whitmore, 1991) modified
Plant	Crop phenology and development	LINTUL2 (van Oijen and Lefelaar, 2008)
	Carbon assimilation, shoot growth and assimilate partitioning	LINTUL2 (van Oijen and Lefelaar, 2008), modified according to Shibu et al. (2010)
	Root growth	SlimRoots (Addiscott and Whitmore, 1991)
	Crop nitrogen demand	LINTUL3 (Shibu et al., 2010)
	Crop water demand	LINTUL2 (van Oijen and Lefelaar, 2008)

average air temperature (Williams and Izaurralde, 2005). Soil water content and water fluxes are calculated with the HILLFLOW sub-model (Bronstert and Plate, 1997), which solves the physically based Darcy equation for unsaturated flows. A special feature of the approach given by Bronstert and Plate (1997) is the consideration of macropore flow, when the intensity of incoming rain exceeds infiltration capacity of the first layer. Soil carbon and nitrogen turnover is simulated according to the SoilCN model developed by Corbeels et al. (2005).

2.3. Modeling biopore effects on root development

In order to take into account the effect of biopores on root growth in heterogeneous soils, the sub-model SlimRoots (Addiscott and Whitmore, 1991, see previous chapter) has been extended. In addition to the effect of soil temperature, penetration of seminal roots is limited by a dimensionless bulk density reduction factor BDRF which reduces the actual growth of the seminal roots in layer L as follows

$$\mathrm{ASROOT(L)} = \mathrm{ASROOT(L)} \times \mathrm{BDRF(L)}$$

where ASROOT is the amount of assimilates supplied to the seminal root in g m^{-2}.

BDRF is calculated for each soil layer by the equation given by Williams and Izaurralde (2005):

$$\mathrm{BDRF} = \frac{\mathrm{BD}}{\mathrm{BD} + e^{(\mathrm{br1}+\mathrm{br2}\times\mathrm{BD})}} \quad (1)$$

where BD is the bulk density (in g cm^{-3}) in the respective layer and br1 and br 2 are dimensionless coefficients which depend on the upper (BDU) and lower (BDU) limit of the soil strength factor. The upper and lower limits are a function of sand content (SAN in %) in each layer and a bulk density threshold for root extension stress in a soil with zero sand content (default threshold BDTH = 1.25 g cm^{-3}) (Jones, 1983). The variables br1 and br2 are calculated according to Eqs. (2) and (3):

$$br2 = \frac{\ln(0.01124 \times \mathrm{BDL}) - \ln(8.0 \times \mathrm{BDU})}{\mathrm{BDL} - \mathrm{BDU}} \quad (2)$$

$$br1 = \ln(0.01124 \times \mathrm{BDL}) - br2 \times \mathrm{BDL} \quad (3)$$

with

$$\mathrm{BDL} = \mathrm{BDTH} + 0.00445 \times \mathrm{SAN} \quad (4)$$

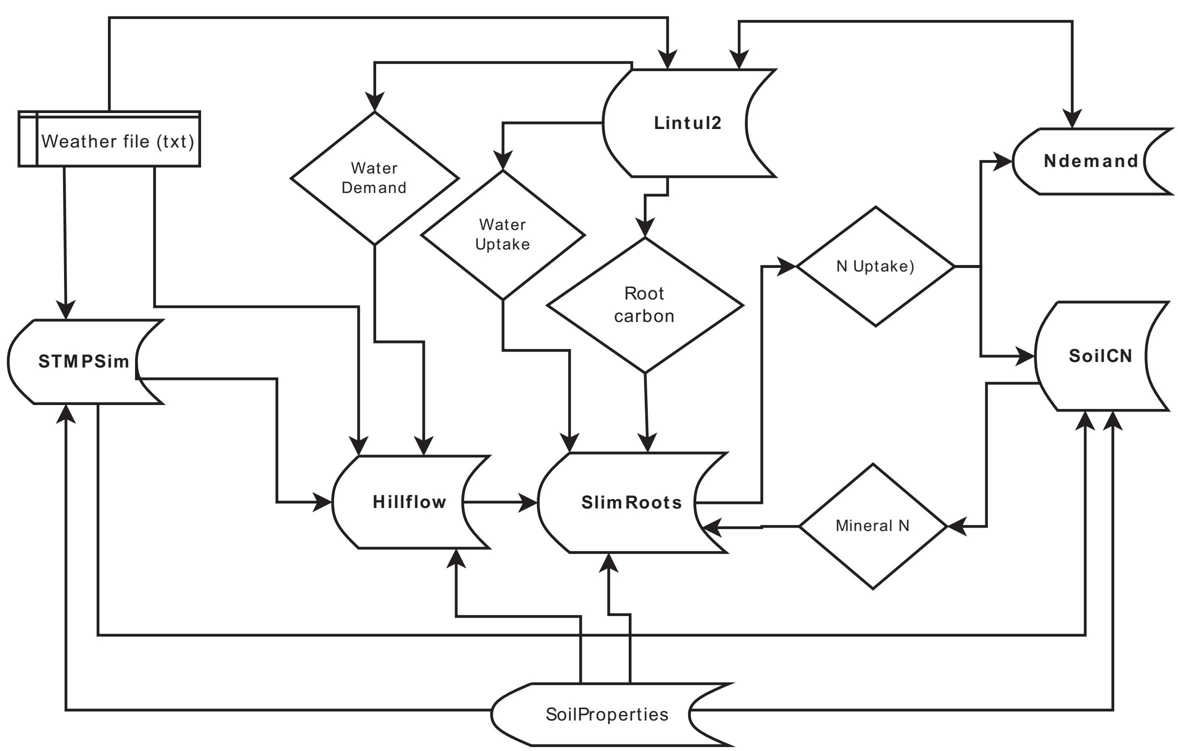

Fig. 1. Scheme of interfaces between sub-models in the solution SIMPLACE<Lintul2,STMPsim,SlimRoots,Hillflow,SoilCN>.

Table 2
Mean bulk density ($g\,cm^{-3}$), proportion of biopore volume (MPV, $m^3\,m^{-3}$), clay and sand content (%) in the topsoil (0–30 cm) and the subsoil (45–90 cm) of a Haplic Luvisol over eight observation plots.

	Bulk density	Clay	Sand	MPV
Topsoil				
Min	1.31	14.6	4.5	0.0038
Max	1.46	19.5	8.8	0.0049
Mean	1.42	17.4	7.2	0.0044
STD	0.1	2.0	1.9	0.0005
Subsoil				
Min	1.52	27.2	3.0	0.0050
Max	1.62	34.2	7.0	0.0083
Mean	1.57	31.0	4.3	0.0062
STD	0.03	2.3	1.2	0.0011

and

$$\mathrm{BDU} = \mathrm{BDTH} + 0.35 + 0.005 \times \mathrm{SAN} \tag{5}$$

Observations at the experimental station have shown that the site was characterized by a high density of biopores (pores with a rounded shape and a diameter > 2 mm). Biopores are facilitating the penetration of roots through dense soil layers and thus counteract the influence of soil strength. The effect of the biopores on dampening the bulk density reduction factor in each soil layer is considered by the following equation:

$$\mathrm{MPSSF} = \min\left(1, \mathrm{BDRF} + \mathrm{MPV} \times \frac{1 - \mathrm{BDRF}}{\mathrm{MPRT}}\right) \tag{6}$$

where MPSSF is the dimensionless biopore-soil strength factor for each layer, MPV is the proportion of biopore volume (in $cm^3\,cm^{-3}$) in a soil layer and MPRT is the proportion of biopore volume (in $cm^3\,cm^{-3}$) above which root penetration is no longer affected by soil strength (called biopore – root growth threshold in the following).

3. Field observations and data analysis

The observations were obtained from a field experiment which was established on a Haplic Luvisol (IUSS, 2006) derived from loess at the agricultural experimental station of the University of Bonn in 2010. Soil properties averaged over topsoil and subsoil are shown in Table 2. The soil was characterized by a silty clay loam texture with clay accumulation in the subsoil (between 45 and 95 cm soil depth). In all plots, a distinct change in texture and increase in bulk density occurred at about 45 cm soil depth, which is typical for the argic horizon of the soil group Luvisols (Table 2). The argic horizon constitutes a strong impediment for root penetration and root extension in particular when dry. The growing season in 2010, was characterized by prolonged dry conditions during the crop cycle of spring wheat. The decadal climatic water balance (calculated as the difference between rainfall and potential evapotranspiration) was zero or negative from April to mid of August except for the first decade in May. Effects of four treatments, representing different cropping sequences with fodder crops on biopore density and root growth of spring wheat, were investigated in a randomized complete block design (Table 3).

Table 3
Treatments and related crop sequences under study.

Year	Luc2Y	Chi2Y	Chi3Y	Fes1Y
2007	Rye	Rye	Chicory	Rye
2008	Lucerne	Chicory	Chicory	Oats
2009	Lucerne	Chicory	Chicory	Fescue
2010	Spring wheat	Spring wheat	Spring wheat	Spring wheat

The soil was tilled with a mouldboard plough to a depth of 30 cm before spring wheat (cultivar: Scirocco) was sown with a density of 450 seeds m^{-2} on April 7th, 2010. In order to realize the same level of plant available nitrogen in all plots, Calcium ammonium nitrate (CAN) was applied with amounts of 15 kg ha^{-1} N (Luc 2Y) and 77 kg ha^{-1} N (Chi2Y, Chi3Y, Fes1Y) according to the amount of available soil nitrogen content in 0–90 cm soil depth in the four treatments. Total above ground biomass of spring wheat was measured on July 13th, 2010 from subplots of 0.25 m^2 within the eight plots with 2 replicates per plot and dry matter content was determined. Crop phenology was monitored regularly from tillering to maturity.

Biopore density was determined in April 2010 in four field replications after sowing spring wheat. Horizontal areas of 50 cm × 50 cm were excavated in 45 cm soil depth and cleaned from soil particles with a vacuum cleaner. Biopores of different diameters (<2 mm, 2–5 mm and >5 mm) were marked on plastic sheets that were placed on the horizontal soil surface. In 2011, biopore observations were extended down to 145 cm in the treatments Chi2Y and Fes1Y in two field replications. As in 2010 the investigation started in 45 °cm soil depth. From 45 to 145 cm biopores were measured in 10 cm-steps as described before. The site was characterized by a relatively dense pattern of biopores partially generated by the different pre-crops (Table 3). The biopore density in the plough layer (0–30 cm) is highly variable within the year due to soil tillage and was assumed to be equal to the density observed at 45 cm. Based on the decrease of biopore density per cm soil depth, vertical distribution of biopore densities in the other two treatments were estimated. Finally, assuming a cylindrical shape and a continuous mean diameter of 4 mm within a soil layer, the volumetric proportion of the biopores in each soil layer was calculated.

Root-length density (RLD) of spring wheat was estimated on June 30th and on July 30th 2010 at the end of flowering and at maturity with the profile wall method (Böhm, 1979; Köpke, 1979). Observations were made in the four treatments under investigation (Table 1) with two field replications. An excavator was used to install a trench (depth: 1.8 m). A 100 cm wide soil profile wall was smoothened to maximum rooting depth and transversely to the plant rows with a spade and sharp blades. Roots exposed from the wall were removed by scissors. With a fine spray of water at 3 bar pressure and the use of a small toothed scraper a soil layer nearly 0.5 cm thick was washed away along the vertical wall of the soil pit. Thus, surface roots were exposed. A frame (inner dimensions 100 cm × 60 cm) with grids (5 cm × 5 cm) was attached to the profile wall. Root length units (RLU) equivalent to 0.5 cm root length were recorded in each square. The root length density from the profile wall (RLD_{PW} in cm cm^{-3} soil) was calculated for each observed square according to the following equation:

$$\mathrm{RLD_{PW}} = \frac{\mathrm{RLU} \times 0.5\ \ \mathrm{cm}}{5\ \ \mathrm{cm} \times 5\ \ \mathrm{cm} \times 0.5\ \ \mathrm{cm}} \tag{7}$$

It is well known, that RLD observations with the profile wall method are useful to assess relative differences between different treatments with respect to their effect of root length density, but the method underestimates the absolute root length density as compared to the destructive sampling of monoliths with a defined volume and the direct measurement of root length (Gäth and Meuser, 1989; Vepraskas and Hoyt, 1988). Therefore, the observations obtained by the profile wall method had to be calibrated against direct measurements from soil monoliths. Thus, a calibration equation was established by fitting the RLD values obtained with the profile wall method against measurements of RLD from 58 monoliths (10 cm × 10 cm × 25 cm with 3 replicates per plot and depth). The monoliths were extracted in the same plots from the succeeding crop (barley) at different intervals within 45–105 cm soil depth. In the topsoil, 16 auger samples (auger diameter 9.3 cm with 2 replicates per plot and depth) were taken at 0–10 cm and 10–20 cm soil depth. The resulting calibration equation was

$$\mathrm{RLD} = \mathrm{RLD}_{\mathrm{PW}} \times (3.0092 + 13.5 \times \mathrm{e}^{-7.2969\times \mathrm{RLD}_{\mathrm{PW}}}) \quad (8)$$

where RLD is the absolute RLD (in cm cm^{-3}) obtained from soil monoliths and auger samples and $\mathrm{RLD}_{\mathrm{PW}}$ is the RLD observed with the profile wall method.

Average root elongation rates of spring wheat were estimated from the maximum rooting depth observed on June15th, 2010 at the profile wall divided by the days after emergence (62 days). The observed and simulated elongation rates in this paper refer to the mean rate of the vertical penetration of seminal roots into the soil and are considered as an average elongation rate over several individual plants (in the case of the root observations with the profile wall method) or over a certain spatial unit (e.g. square meter or hectare in the simulations).

4. Initial model parameters of simulation runs

After calibrating the soil temperature sub-module as well as the crop phenology sub-module with continuous observations carried out in 2010, root length density (RLD) of spring wheat in 2010 was simulated for eight intensive monitoring plots covering four treatments with two replicates each (Luc2Y = 2 years lucerne, Chi2Y = 2 years chicory, Chi3Y = 3 years chicory, Fes1Y = 1 year fescue). Initial soil properties down to 105 cm were determined either plot wise (soil bulk density, saturated hydraulic conductivity) or treatment wise (biopore density, water retention curve, SOC and SON content). For the soil layer between 105 and 150 cm soil property data from the soil survey by Rehkopf et al. (unpublished) were used as initial parameters for the model runs. The soil water balance model HILLFLOW considered biopore flow down to 1.5 m and ponding of water within the pores (Table 4). Below the maximum root depth of 1.5 m free-flow conditions were assumed.

Model runs were carried out for MPRT values ranging from 0 to 0.032 m^3 m^{-3} and the simulated RLDs down to 80 and 110 cm were compared with the observed RLDs at two observation dates carried out on June 30th and July 30th 2010, respectively.

5. Data analysis

As quality indicators for the performance of the modeling approach the following parameters were used (Loague and Green, 1991; Smith et al., 1996):

a. coefficient of determination R^2, slope and intercept of the linear regression between observed and simulated values
b. mean residual error ME as

$$\mathrm{ME} = \frac{1}{n}\sum_{i=1}^{n}(y_i - x_i)$$

c. mean relative error MRE as

$$\mathrm{MRE} = \frac{1}{n}\sum_{i=1}^{n}\frac{(y_i - x_i)}{x_i}$$

d. mean absolute error MAE as

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}(|y_i - x_i|)$$

e. model efficiency as

$$\mathrm{EF} = \left[\sum_{i=1}^{n}(x_i - \bar{x})^2 - \sum_{i=1}^{n}(y_i - x_i)^2\right] / \sum_{i=1}^{n}(x_i - \bar{x})^2$$

f. root mean square error

$$\mathrm{RMSE} = \left[\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i)^2\right]^{0.5} \times \frac{100}{\bar{x}}$$

where n is the sample number, x is the observed, y is the simulated value and $\bar{x}$ is the mean of the observed values. Values of mean residual and mean relative error close to zero indicate small differences and little systematic deviation between simulated and measured values. The same conclusion can be drawn if coefficient of determination, slope of the linear regression or model efficiency are close to one. Values of absolute error or root mean square error close to zero rather express precision and reliability of the prediction for single yield estimation points.

6. Results

6.1. Biopore density

Biopore density in the four treatments was in the range of 300–390 m^{-2} at 45 cm soil depth. Assuming a continuous mean diameter of 4 mm within a soil layer. This corresponds to a volumetric proportion of 0.0038–0.0049 m^3 m^{-3} depending on the soil layer and treatment (Table 2). At the time of biopore density quantification, many fodder crop tap roots were not fully decayed, still blocking the pores they created or colonized. Thus, the data in Table 2 does not represent the long-term biopore density affected by the four pre-crop treatments but the situation at the beginning of the spring wheat growing cycle in April 2010. Over the whole soil profile, the biopore density was between 300 and 660 m^{-2} corresponding to a volumetric proportion of 0.0038–0.083 m^3 m^{-3} (Table 2). Biopore density increased with depth in the argic horizon with a maximum at 45–55 cm and then decreases steadily. At 145 cm soil depth the proportion of biopores was still above 0.3% per volume (corresponding to approximately 300 m^{-2}) (data not shown).

6.2. Biopore effects on root elongation rates as affected by the a biopore-root growth threshold

The new modeling approach proposed here assumes a linear relationship between the biopore-soil strength factor (MPSSF) and the proportion of biopores per volume of soil with a given threshold value (MPRT = biopore root growth threshold), where the density of biopores off-sets the effect of soil strength on root penetration. The effect of MPRT on the biopore-soil strength factor is shown in Fig. 2, assuming a given bulk density restriction MPSSF of 0.5 when no biopores are present. The proposed approach renders root penetration rate sensitive to changes in the proportion of biopore volume and to changes in bulk density (Fig. 3).

Fig. 3 shows that maximum root elongation (3.3 cm d^{-1}) is hardly affected by bulk densities below 1.4 g cm^{-3} regardless of the biopore-root growth threshold (MPRT). However, if bulk density is above 1.4 g cm^{-3} root elongation rate decreases sharply, if no biopores are present. This reduction in elongation rate through bulk density is alleviated by increasing biopore volume until the maximum root elongation rate is reached, indicated by the plateau at

Table 4
Initial parameters for the submodels Lintul2, SlimRoots and Hillflow.

Parameter	Description (unit)	Default value
Lintul2/Lintul3		
LUE	Light use efficiency ($g\,m^{-2}\,MJ^{-1}$)	4.5
SLA	Specific leaf area ($g\,m^{-2}$)	0.035
RGRL	Relative growth rate of LAI in the early development stage (–)	0.018
BaseTempAnthesis	Base temperature before anthesis (°C)	0
BaseTempAfterAnthesis	Base temperature after anthesis (°C)	0
TsumMaturity	Growing degree days until maturity (°C d)	1100
Drought tolerance	Drought tolerance factor (–)	1.0
NoptimalFraction	Fraction of N concentration below which nitrogen stress occurs (–)	0.85
HILLFLOW		
Hlim3l	Lower threshold of soil hydraulic potential at which crop transpiration is reduced (cm WC)	−100
Hlim4	Threshold of soil hydraulic potential at which crop transpiration is zero (cm WC)	−1000
DZF	Depth of macropore flow (m)	1.5
SlimRoots		
MaximumRootElongation Rate	Maximum elongation rate of seminal roots per day ($cm\,d^{-1}$)	3.3
FLOWNR	Maximum N uptake capacity by roots ($mg\,N\,m^{-1}$)	40
SoilStrengthParameter	Bulk density threshold above which root elongation is affected by soil strength ($g\,cm^{-3}$)	1.25
DailyDeadRootsStage	Development stage at which root length density starts to decrease	1.2

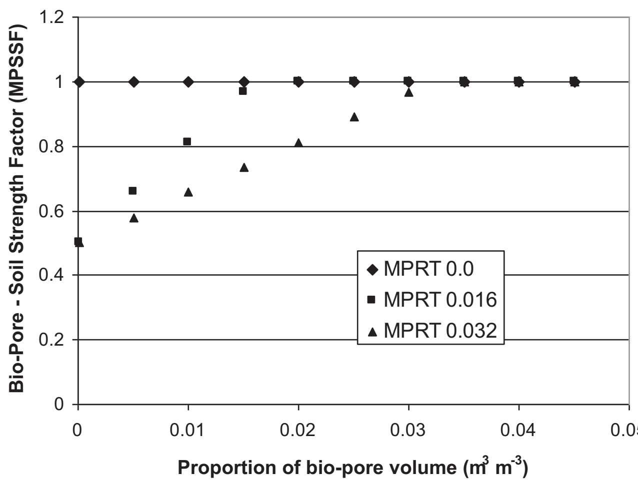

Fig. 2. Relationship between the proportion of biopore volume and the biopore soil strength factor (MPSSF) in a soil where MPSSF without biopores is 0.5 as a function of the biopore-root growth threshold value MPRT (threshold value where biopores volume offsets the effect of soil strength on root penetration).

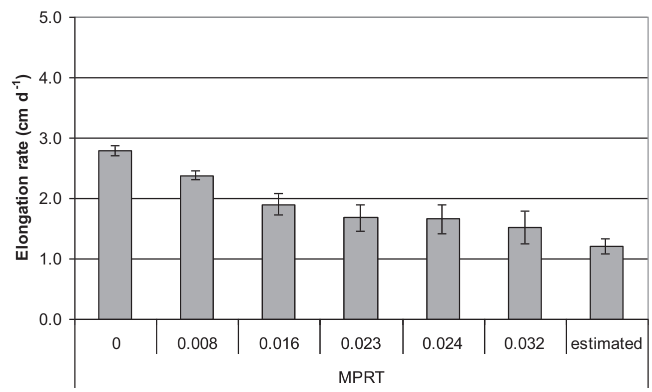

Fig. 4. Simulated mean root elongation rates ($cm\,d^{-1}$) of spring wheat when reaching maximum rooting depth (between June 14th and June 26th) as related to the root growth-biopore threshold (MPRT) compared to mean estimated elongation rates on June 15th (flowering).

$3.3\,cm\,d^{-1}$ in Fig. 3. The lower the MPRT value the faster this plateau is reached with increasing biopore volume.

6.3. Model calibration using root observations at the plot scale

Although several investigations have studied the effect of bio- or macro-pores on root growth and root penetration rates of wheat and maize (Jakobsen and Dexter, 1987; Valentine et al., 2012; Volkmar, 1996), none of them has attempted to quantify a threshold value for the biopore volume where biopores allow free penetration of roots in spite of soil structural barriers. Thus, the root observations carried out in the four treatments of the field experiment with contrasting biopore density offered an excellent opportunity to optimize the MPRT parameter at the plot scale through inverse modeling.

The simulated mean elongation rates of spring wheat over the entire growing period were in the range of $2.8–1.5\,cm\,d^{-1}$ with the maximum elongation rate being set to $3.3\,cm\,d^{-1}$ (Fig. 4). With

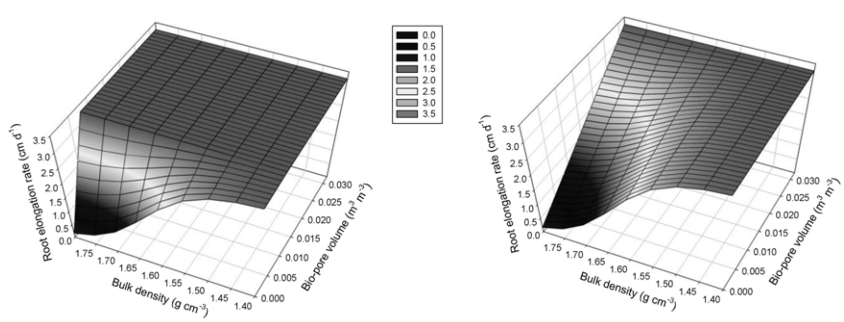

Fig. 3. Root elongation rate ($cm\,d^{-1}$) as affected by bulk density ($g\,cm^{-3}$) and biopore volume ($m^3\,m^{-3}$) for biopore root growth threshold MPRT = $0.008\,m^3\,m^{-3}$ (left) and MPRT = $0.032\,m^3\,m^{-3}$ (right) (assumed maximum elongation rate $3.3\,cm\,d^{-1}$, default bulk density threshold and sand content of 30%).

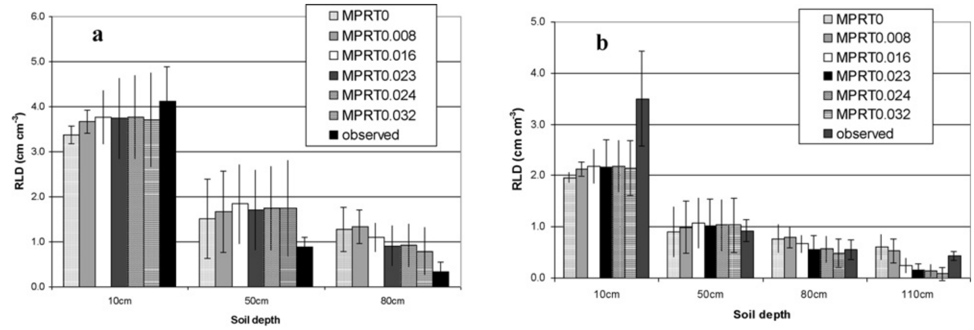

Fig. 5. Simulated root length densities of spring wheat in different soil depths as related to the MPRT value compared to observed root length densities on (a) June 30th (flowering) and (b) July 30th (maturity).

increasing threshold values, the mean elongation rates dropped to 1.5 cm d^{-1} when MPRT was 0.032 m^3 m^{-3}. This was close to the estimated average root elongation in the eight monitoring plots, which amounted to 1.2 cm d^{-1}. At flowering, simulated RLD slightly increased with increasing MPRT at 10 cm soil depth until a maximum was reached with MPRT of 0.032 m^3 m^{-3} (Fig. 5a). In the subsoil (80 cm) RLD decreased continuously with MPRT values above 0.008 m^3 m^{-3}. At maturity, simulated RLD in the topsoil was much lower than observed, but in 50 cm and in particular in 80 cm depth, simulated RLD corresponded well to observed RLD when MPRTs of 0.023 m^3 m^{-3} or 0.024 m^3 m^{-3} were used (Fig. 5b). At 110 cm soil depth, the observed RLD was almost as high as in 80 cm depth.

As the vertical distribution of RLD is the most critical factor which influences water and nutrient uptake and consequently biomass production by the crop, the model parameter MPRT was optimized so as to determine the best fit between simulated and observed RLD over the soil profile. The model performance indicators root mean squared error (RMSE) and model efficiency were selected as target indicators for the parameter optimization procedure. The root mean squared error (RMSE) reached its minimum at a MPRT of 0.02307 m^3 m^{-3} whereas model efficiency EF was found to be highest at a MPRT = 0.02311 m^3 m^{-3} (Fig. 6). Thus, a biopore volume threshold of 0.023 m^3 m^{-3} was considered to be best suited to reproduce RLD distribution at this site. This was supported by the fact that the simulated mean biomass production and in particular the simulated standard deviation were close to the observed values (Fig. 7). Therefore, a MPRT of 0.023 m^3 m^{-3} was used to assess drought stressed spring wheat biomass production under scenarios of maximum biopore volume (biopore volume equal to MPRT) and zero biopore volume from top- to sub-soil or zero biopore volume in the subsoil only.

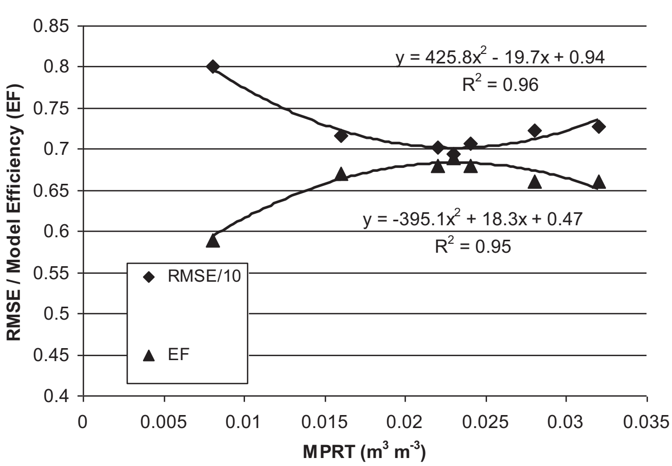

Fig. 6. Parameter optimization for MPRT using model efficiency (EF) and root mean squared error (RMSE in m^3 m^{-3}) as performance indicators for the best fit between simulated and observed root length density at maturity (N = 160), for both indicators the optimum is reached at MPRT = 0.0231 m^3 m^{-3}.

6.4. Sensitivity analysis to evaluate the effect of biopores on root growth and biomass production of drought stressed spring wheat

After parameter optimization for MPRT the model was used to explore the effect of different proportions of biopores in the soil at the experimental station on drought stressed spring wheat. A total of four scenarios were evaluated: (1) proportion of biopore volume over all soil layers is equal to MPRT (0.024 m^3 m^{-3}), (2) proportion of biopore volume corresponds to the measured values in each soil layer, (3) proportion of biopore volume in the subsoil layers (>45 cm soil depth) is equal to zero, (4) proportion of biopore volume over all soil layers is equal to zero. Fig. 8 shows that a decreasing proportion of biopore volume affected simulated mean root elongation rate of the spring wheat stand negatively. If no biopores were present (MPV0), the mean elongation rate was reduced to less than 50% compared to a soil with a volumetric proportion of biopores of 0.024 m^3 m^{-3} throughout the whole soil profile (MPVmax). The reduced simulated root elongation rate in the scenario without biopores (MPV0) negatively affected water and nitrogen uptake by spring wheat. In the case of no biopores (MPV0), simulated water uptake decreased by 44 mm (−16%) and simulated nitrogen uptake decreased by 45 kg ha^{-1} (−27%) compared to a soil with a proportion of biopore volume of 0.023 m^3 m^{-3} (MPVmax) (Table 6). The reduction in root elongation rate, water and nitrogen uptake finally translated into a decrease in biomass production

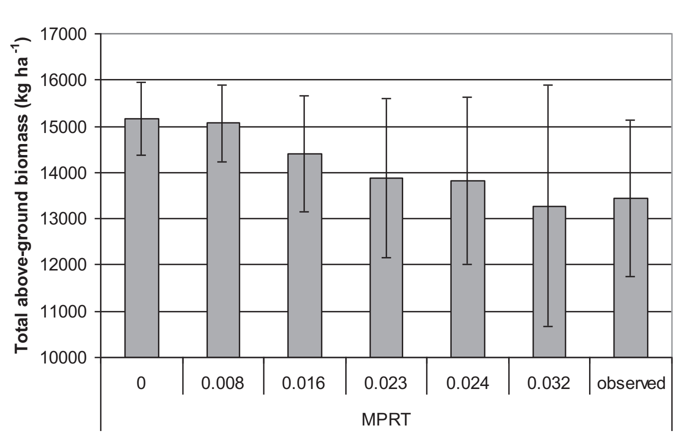

Fig. 7. Simulated total above ground biomass production by spring wheat at grain filling under drought stress as related to the MPRT value compared to observed biomass production.

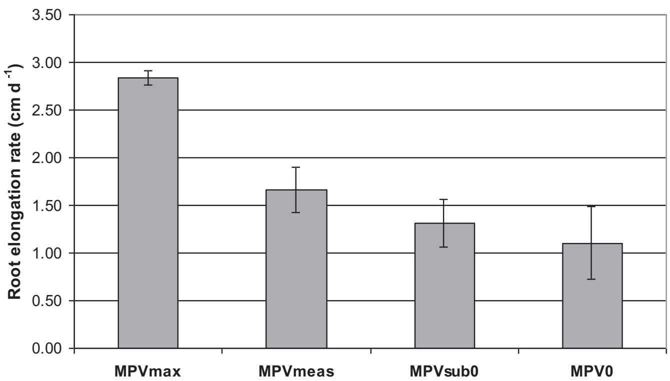

Fig. 8. Simulated mean root elongation rates by drought stressed spring wheat as affected by varying proportions of biopores in the soil (MPVmax, biopore volume is equal to 0.023 $m^3 m^{-3}$ in all soil layers; MPVmeas, measured biopore volume in all soil layers (Table 1); MPV0sub, zero biopore volume in the subsoil; MPV0, zero biopore volume in the whole soil profile).

when biopores are missing. Table 6 demonstrates the importance of biopores for the biomass production of drought stressed spring wheat on soils with clay accumulation. The difference in biomass production between a soil with large proportion of biopores (MPVmax) and without biopores (MPV0) was more than 4000 kg ha^{-1}, which equals to a decrease of about 26% compared to MPVmax.

7. Discussion

7.1. Biopore effects on root elongation rates and root distribution

The potential or maximum elongation rates of wheat found in the literature are in the order of 3.3 cm d^{-1} (Bingham and Wu, 2011; Pritchard et al., 1987). At our site the mean root elongation rate observed over the period April to June (1.2 cm d^{-1}) was much lower than the maximum elongation rates, confirming that root penetration into the subsoil was negatively affected by the clayey soil layers starting at 45 cm. The new modeling approach proposed here renders root penetration rates of wheat sensitive to changes in soil bulk density and the degree of sensitivity depends on the volume of biopores (Fig. 3). Such interactions have been observed in several investigations studying the effect of biopores on root growth and root penetration rates in soils with subsoil compaction (Jakobsen and Dexter, 1987; Nakamoto, 1997; Valentine et al., 2012; Volkmar, 1996).

This paper presents an attempt to simulate the effects of biopores on root penetration at the plot scale and to quantify the volume of biopores in the soil (MPRT = biopore-root growth threshold) which is necessary to attain maximum rates of root elongation regardless of the soil density. As the mean measured proportion of biopore volume at out site was 0.0062 $m^3 m^{-3}$, MPRT values above 0.0062 $m^3 m^{-3}$ negatively affected simulated root elongation rate and vertical root distribution, whereas with MPRT values below 0.0062 $m^3 m^{-3}$ the simulated root elongation rates were only temperature limited (Fig. 4). Therefore, with an MPRT value of 0 the simulated root elongation rate of 2.8 cm d^{-1} was closest to the maximum root elongation rate of 3.3 cm d^{-1}. As shown in Figs. 4 and 8a decreasing proportion of biopore volume negatively affected simulated mean root elongation rate of spring wheat. Valentine et al. (2012) showed in soil core trials with barley seedlings, that root elongation was strongly linked to differences in physical soil properties. In the majority of field soils, the root elongation rate was slower than half of the unimpeded rate. The higher MPRT, the smaller was the simulated root elongation rate, reaching 1.4 cm d^{-1} with MPRT of 0.032 $m^3 m^{-3}$. However, the average observed root elongation rate was likely to be underestimated, because the profile wall method tends to underestimate RLD (Gregory, 2006). In particular as the root system is expanding into the subsoil, where RLD is very low, there is a high probability that the deepest roots at the fringe of the expanding zone are not detected by this method. Thus, the real average elongation rate was most likely higher than 1.2 cm d^{-1} and within the range of simulated elongation rates found at MPRT values of 0.023 and 0.024 $m^3 m^{-3}$.

The model was optimized by comparing the simulated distribution of RLD along the soil profile with observed root distribution down to 110 cm soil depth. In general, the simulated RLD in the topsoil (10 cm) increased slightly with increasing MPRT (Fig. 5). The increase in simulated RLD in the topsoil with increasing MPRT was due to the investment of assimilates into the root system of the topsoil if root penetration into deeper layers is impeded. This has been observed with small grain cereals like wheat as in our case, but it is not a universal mechanism (Crossett et al., 1975; Kutschera et al., 2009). In the deeper subsoil layers (below 50 cm) in particular at 110 cm soil depth, the observed RLD at maturity was almost as high as in 80 cm depth. This was probably due to the fact that spring wheat was under drought stress during flowering and grain filling and the crop invested more assimilates into the deeper roots. This reaction of herbaceous crops under drought stress has been observed in field studies with maize and wheat (Gregory, 2006; Sharp and Davies, 1985), but is not yet implemented in the current SIMPLACE solution. This may be one of the reasons for the high mean relative error (MRE) as well as RMSE of the simulated RLDs even with the optimal MPRT of 0.023 $m^3 m^{-3}$ (Table 5). At a given bulk density and with increasing biopore volume the root elongation rate increased, which caused an increasing RLD in deeper layers in the simulations. In a soil core experiment taken from a Dark Brown Chernozemic soil, Volkmar (1996) observed that the density of large pores correlated positively with the number of wheat roots thus confirming the simulated effects of biopores on root elongation rates and root density.

7.2. Biopore effects on biomass production of drought stressed spring wheat

It is obvious that a reduced root density in the subsoil translates under non-optimal water and nutrient supply into reduced

Table 5
Model performance indicators for simulated versus observed root length density (cm cm^{-3}) at maturity (N = 160) (ME, mean residual error; MRE, mean relative error; MAE, mean absolute error; EF, model efficiency; RMSE, root mean squared error; R^2, coefficient of determination).

MPRT ($m^3 m^{-3}$)	ME	MRE	MAE	EF	RMSE	R^2	Slope	Intercept
MPRT 0	0.19	1.12	0.73	0.54	84.3	0.57	1.15	−0.38
MPRT 0.008	0.26	1.26	0.70	0.59	80.0	0.63	1.08	−0.38
MPRT 0.016	0.17	1.11	0.58	0.67	71.7	0.68	0.99	−0.17
MPRT 0.022	0.10	0.97	0.54	0.68	70.3	0.69	0.95	−0.04
MPRT 0.023	0.08	0.92	0.54	0.69	69.4	0.69	0.95	−0.02
MPRT 0.024	0.09	0.95	0.55	0.68	70.7	0.69	0.93	−0.01
MPRT 0.028	0.06	0.90	0.56	0.66	72.4	0.67	0.91	0.05
MPRT 0.032	0.04	0.83	0.56	0.66	72.8	0.67	0.89	0.08

Table 6
Simulated above ground biomass (ABG in kg ha^{-1}), total water and nitrogen uptake by drought stressed spring wheat as affected by varying proportions of biopores in the soil (MPVmax, biopore volume is equal to 0.023 m^3 m^{-3} in all soil layers; MPVmeas, measured biopore volume in all soil layers (Table1); MPV0sub, zero biopore volume in the subsoil; MPV0, zero biopore volume in the whole soil profile).

	ABG (kg ha^{-1})	Water uptake (mm)	N uptake (kg ha^{-1})
MPVmax	15,162	284	168
MPVmeas	13,886	277	150
MPV0sub	12,751	263	139
MPV0	11,156	240	124

uptake rates as demonstrated by the simulation results (Table 6). Volkmar (1996) found that the root number of wheat was enhanced by large pores and that the number of roots was positively correlated to the shoot N content. In our biopore scenarios, the highest simulated water and nitrogen uptake in the soil associated with the highest proportion of biopores (MPVmax) was mostly likely due to the improved exploitation of humid subsoil layers which are reached faster when root elongation rates are higher (Fig. 8). The importance of a fast root development is particularly important for spring crops because they have a shorter growing cycle (in Germany 120–160 days in contrast to winter crops with more than 200 days) and thus have less time to reach subsoil layers rewetted by the winter rains.

The lower simulated biomass production in the soil without biopores (MPV0) compared to the soil with large proportion of biopores (MPVmax) was then caused by the smaller water and nutrient uptake in the former (Table 6). This relationship between biopore densities and wheat productivity is in agreement with simulation results by Jakobsen and Dexter (1987) who found yield reductions of spring wheat in Southern Australia between 10 and 30% in a dense soil when no macropores (cracks) were present. They explained the yield reduction in the soil with no macropores by the increased mechanical resistance and the resulting poor root penetration by wheat. In our investigations, the difference between the yield in the soil without biopores and the soil with observed biopore volume was only 2730 kg ha^{-1} (−20%), but illustrates the importance of biopores for production of short-cycle spring wheat at this site (Table 6). This effect may be less pronounced in wheat with a longer crop-cycle or when drought stress is less pronounced as in this particular year. But when spring crops on such soils are under drought stress, a rapid penetration of the roots into deeper humid soil layers seems to be critical for maintaining maximum biomass production and yield.

8. Conclusions

Soils with clay accumulation and significant increase of soil strength in the subsoil cover a considerable proportion of the global land surface, but they bear the risk of root barriers, impeding fast penetration of sub-soil layers. Biopores may constitute channels for vertical roots to overcome the physical root barrier more rapidly. This paper presents a modeling approach to describe such interactions at the plot scale. The modeling approach was embedded into the SIMPLACE framework and then calibrated against root observations from a field experimental with a wide range of biopore densities. The calibrated model was used to perform a sensitivity analysis, which demonstrated the importance of biopores to increase the penetration rate of roots through layers with higher soil strength to rapidly reach deeper, more humid subsoil layers. This is crucial for biomass production of short-cycle spring wheat when prolonged dry spells occur. The model, embedded into SIMPLACE, allowed a rough quantification of the biopore effects with respect to root elongation rate and biomass production at the plot scale. Further investigations need to clarify to which extent these findings are also applicable at to other crops with longer growing cycle.

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